

Day - 1: ARM Basics

▼ Microprocessor vs Microcontroller

1. Microprocessors (Example: 8086)

A microprocessor is an integrated circuit that functions as a standalone CPU.

- **Core Components:** It contains an Arithmetic Logic Unit (ALU), general-purpose registers, and control units.
- **Missing Components:** It does **not** have built-in memory, input/output (I/O) ports, or serial/parallel communication support. These must be connected externally.
- **8086 Specifications:**
 - It is a 16-bit processor, meaning its ALU handles 16 bits of data at a time.
 - It has a 16-bit data bus and a 20-bit address bus.
 - Operating speed is typically 5 to 10 MHz, though some can operate up to 20 MHz.
- **Key Registers in the 8086:**
 - **General Purpose Registers (AX, BX, CX, DX):** Used for standard arithmetic operations like addition and multiplication.
 - **Pointers (SP, BP):** The Stack Pointer (SP) points to the top of the stack, and the Base Pointer (BP) points to the base.
 - **Index Registers (SI, DI):** Source Index (SI) and Destination Index (DI) act as pointers used to transfer data between memory locations.
 - **Segment Registers (CS, SS, DS, ES):** Code Segment, Stack Segment, Data Segment, and Extra Segment (for strings) are used to point to different sections of externally interfaced memory.
- **Control Unit:** Generates signals such as Read (RD), Write (WR), Memory (MEM), and I/O (IOS).

2. Microcontrollers (Example: 8051)

A microcontroller acts as the "heart" of an embedded system, while the processor inside it acts as the "brain".

- **Core Components:** Unlike a microprocessor, a microcontroller includes a CPU, memory (RAM and ROM), I/O ports, timers, and communication ports all packaged onto a single chip.
- **8051 Specifications:**
 - It is an 8-bit microcontroller (8-bit CPU and 8-bit data bus) with a 16-bit address bus.
 - Features 4 KB of ROM and 120 bytes of RAM.
 - Contains a Program Status Word (PSW) status register and 8 general-purpose registers (R0 to R7).
- **Timers & Interrupts:**
 - Includes two 16-bit timers (Timer 0 and Timer 1) used to calculate delays and generate interrupts.
 - Supports 5 interrupt sources: two external, two from timers, and one serial.
- **I/O and Communication:**
 - Features 32 General Purpose Input Output (GPIO) pins, divided evenly across 4 ports (Port 0, 1, 2, and 3).
 - Includes a **UART** (Universal Asynchronous Receiver and Transmitter) for serial communication between devices.

3. Key Differences: Microprocessor vs. Microcontroller

Feature	Microprocessor	Microcontroller
Architecture	Standalone CPU; requires external components (RAM, ROM, timers, I/O).	"System on a chip"; CPU, memory, timers, and I/O are all built-in.
Primary Function	Focuses heavily on data processing.	Focuses on data processing <i>plus</i> controlling and interacting with peripherals.

Feature	Microprocessor	Microcontroller
Instruction Level	Instructions target data processing.	Instructions frequently operate at the bit level for controlling hardware.
Hardware Complexity	Requires many external components to build a system.	Requires minimal external components.
Applications	Used in general computers.	Used in application-specific embedded systems and domestic appliances.
Cost & Power	Higher cost and higher power consumption due to external interfacing.	Lower cost and lower power consumption due to built-in circuits.

▼ Embedded System Basics | Device Driver | Application Layer

1. Embedded Systems Overview

- **Definition:** An embedded system is a branch of electronics engineering that combines both hardware and software to perform specific, dedicated functions.
- **Core Components:** Typically includes a microprocessor or microcontroller, memory, input/output (I/O) devices, and software components.

2. Hardware Architecture & Data Flow

The hardware works sequentially to interact with the physical world:

- **Sensors:** Electronic circuits that detect and capture analog information from the physical world (e.g., mobile touch screens, temperature sensors, or pulse sensors in smartwatches).
- **ADC (Analog to Digital Converter):** Takes the analog signals from the sensors and converts them into digital data so the processor can understand them.
- **Processor & Memory:** The processor acts as the main circuit, calculating and manipulating the digital data based on programming. The memory

stores the code (instructions) and temporarily stores the data being processed.

- **DAC (Digital to Analog Converter):** If the output requires an analog signal (like a mobile phone speaker or television display), the DAC converts the processed digital data back into an analog form.
- **Actuators:** Mechanical devices that take the final processor or DAC input and produce a physical movement (e.g., a motor spinning in a washing machine).

3. Software Architecture

Embedded software is broadly classified into four main layers:

- **Application Software:** Programs written specifically for the end application's requirements, such as a code designed solely to rotate a washing machine motor or display its speed.
- **Middleware (Middle Level Software):** Acts as a necessary bridge connecting the application software with the device drivers.
- **RTOS (Real-Time Operating System):** Manages all hardware and software components. If there are multiple applications running, the RTOS uses logical mechanisms and timing to control their execution order.
- **Device Drivers:** Code exclusively written to directly control specific hardware components, such as memory, GPIOs, or serial ports.
- **Development Tools:** Embedded software is predominantly developed using **C and C++ programming languages**. Developers use compilers and IDEs (Integrated Development Environments) to write and compile the code.

4. Processors vs. Microcontrollers

The video highlights distinct differences between the two main "brains" of an embedded system:

- **Processor:** A clock-driven programmable integrated circuit (IC) that contains an Arithmetic Logic Unit (ALU), a register set, control signals, and address/data buses. **Crucially, a processor has no built-in memory (RAM or ROM), no I/O circuits, and no communication support.** All of these must be connected externally.

- **Microcontroller:** A single IC that bundles a processor core (CPU) alongside essential built-in components. It includes on-chip memory (ROM for code, RAM for data), GPIO (General Purpose Input/Output) pins for external interactions, serial ports for communication, and timers to generate time delays.

▼ ARM Architecture | ARM Processor | Advanced RISC Machine

1. Core Hardware Components

- **ALU (Arithmetic Logic Unit):** The processor features a 32-bit ALU for performing operations.
- **Data Bus:** A 32-bit bidirectional bus used to carry data.
- **Address Bus:** A 32-bit unidirectional bus used to carry memory addresses.
- **Address Register:** A 32-bit register connected to the address bus that points to the specific memory location for reading or writing data.
- **Address Incrementer:** Automatically increments memory locations. It increments by +4 (4 bytes) when operating in **ARM mode** and by +2 (2 bytes) when operating in **Thumb mode**.
- **Instruction Decoder:** A circuit used to decode the fetched opcode from memory.
- **Sign Extension:** Used to extend data with zeros or ones based on the sign of the number.

2. Bus Multiplexing & Control Signals

The 32-bit Address Bus and 32-bit Data Bus are multiplexed together, meaning only one bus can be active at a time. This is managed using specific control signals:

- **ALE (Address Latch Enable):** If ALE = 0, the data bus is selected; if ALE = 1, the address bus is selected.
- **ABE & DBE:** Address Bus Enable (ABE) and Data Bus Enable (DBE) are used to explicitly enable their respective buses.

- **EN IN & EN OUT:** EN IN (Input Enable) is set to 1 to read data, while EN OUT (Output Enable) is set to 1 to send data from the processor to an external device.

3. Pipelining

The ARM architecture follows a parallel **three-stage pipelining** process:

1. **Fetch:** Retrieves the instruction.
2. **Decode:** Interprets the instruction.
3. **Execute:** Performs the operation. These three stages execute parallelly to improve processor efficiency.

4. Register Organization

The ARM processor contains **16 primary 32-bit registers** (R0 to R15) and one 32-bit status register.

- **R0 to R12 (General Purpose Registers):** Used for standard data operations and general purposes.
- **R13 (Stack Pointer - SP):** Points to the top of the stack for memory management during program jumps.
- **R14 (Link Register - LR):** Stores the return address when jumping from one function or program part to another.
- **R15 (Program Counter - PC):** Acts as a counter that points to the next instruction to be executed.

5. CPSR (Current Program Status Register)

The CPSR is a dedicated 32-bit status register that tracks condition flags and controls processor modes.

- **Condition Flags:**
 - **N (Negative):** Sets if the result of an operation is negative.
 - **Z (Zero):** Sets if the result is exactly zero.
 - **C (Carry):** Sets if a carry is generated from the Most Significant Bit (MSB) or if there is a borrow during subtraction.
 - **V (Overflow):** Sets if the result overflows the specified data range.

- **Control Bits:**
 - **Bit 5 (T Bit):** Selects between operating in **Thumb mode** or **ARM mode**.
 - **Bit 6 (F Bit):** Enables **FIQ (Fast Interrupt Request)** for high-priority interrupts.
 - **Bit 7:** Enables interrupts with moderate/normal priority.
 - **Mode Bits:** The remaining 5 bits are reserved to switch between the 7 different operating modes of the ARM processor.

6. Specialized Hardware Units

- **MAC (Multiply and Accumulate) Unit:** A dedicated multiplier that multiplies two operands (A and B) and adds the result to an accumulator variable (Formula: $\text{Accumulator} = \text{Accumulator} + A * B$).
- **Barrel Shifter:** A specialized 32-bit circuit distinct from the ALU that is exclusively designed to shift or rotate data (e.g., Logical Shift Right, Arithmetic Shift). It is used to shift the second operand by a specified number of bits before the main operation is executed.

▼ What is ARM | ARM Processor | ARM Architecture

1. Introduction to ARM

- **Definition:** ARM stands for **Advanced RISC Machine** (originally known as Acorn RISC Machine).
- **Architecture Type:** It is based on a **RISC** (Reduced Instruction Set Computer) architecture, meaning it utilizes simple and fewer instructions compared to complex processors.
- **Origin:** ARM is designed by a British-based semiconductor company located in Cambridge, England, which started producing processors in 1985.
- **Business Model:** Unlike other semiconductor manufacturers, the **ARM company does not manufacture its own microcontrollers**. Instead, it designs 32-bit and 64-bit architectures and sells licenses to other

companies (such as Atmel, NXP, Samsung, and Texas Instruments) so they can build their own microcontrollers and System-on-Chips (SoCs).

2. Key Processor Features & Hardware Specs

- **Core Configuration:** It is primarily a 32-bit processor, utilizing a 32-bit ALU (Arithmetic Logic Unit), 32-bit registers, and majority 32-bit instructions.
- **Registers:** The processor contains 16 primary registers, labeled R0 through R15.
- **Clock Speed:** Operates at high speeds, typically between **60 MHz to 100 MHz or more**, which is significantly faster than regular microprocessors that usually max out around 30 MHz.
- **Buses:** Features a 32-bit address bus and a 32-bit data bus. These buses are **multiplexed** (unified to save pins) and are demultiplexed using an **ALE** (Address Latch Enable) signal.
- **Memory Range:** It has a 32-bit memory addressing space, allowing the processor to interface with a massive **4 GB of external memory** (232 addressable bytes).
- **Debugging Interface:** Utilizes a **JTAG** (Joint Test Action Group) interface, which allows developers to easily debug the hardware circuit.

3. Pipelining Architecture

To improve speed and efficiency, the ARM processor uses a **three-stage pipelining mechanism**, executing all three stages in parallel:

1. **Fetch:** The processor retrieves the specific opcode (hexadecimal instruction data) from the memory.
2. **Decode:** The fetched opcode passes through a decoder circuit so the processor understands what action to take.
3. **Execute:** The instruction is carried out by the ALU.

4. Operating Modes

The ARM processor supports **7 standard operating modes** and **1 special mode**:

- **Standard Modes:** User, FIQ (Fast Interrupt Request), IRQ (Interrupt Request), Supervisory (SVC), Undefined, Abort, and System mode.
- **Thumb Mode (Special Mode):** This is a unique **16-bit operating mode**. It is primarily used to reduce the overall program size and conserve memory in systems where space is constrained.

5. Advantages & Applications

- **Advantages:** ARM processors are highly accurate, extremely lightweight, low-cost, and consume very little power.
- **Applications:** Because of its efficiency, ARM architecture is heavily used in mobile phones (like iPhones), tablets, laptops, desktop servers, and countless embedded system applications.